

Supplementary Material for “A Semantic Mutation Metric for Metamorphic-Relation Adequacy in Scientific Computing Programs”

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CCS Concepts: • **Software and its engineering** → **Software testing and debugging**; *Software verification and validation*.

Additional Key Words and Phrases: metamorphic testing, mutation testing, semantic mutation testing, metamorphic relation adequacy, metamorphic relation assessment, scientific computing

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A Notation and Operator Catalogue

A.1 Complete notation table

Experimental objects.

- PUT set $I = \{A1, A2, A3, B1, B2, B3, C1, C2, C3, D1, D2, D3\}$, $|I| = 12$.
- PUT: S_i ($i \in I$).
- Class mapping: $\text{cls} : I \rightarrow \{A, B, C, D\}$ (open, extensible).
- Valid input distribution: $\mathcal{D}_P$, sampling $X_{K_{\text{eq}}} \sim \mathcal{D}_P$.

Metamorphic testing side.

- Metamorphic-relation label set: $\mathcal{R} = \{\text{Cons.}, \text{Mono.}, \text{Conv.}, \text{Traj.}, \text{P-ord.}\}$, $k \in \{1, \dots, 5\}$ (open, extensible).
- Metamorphic Relation: $\text{MR}_{i,k}$ (from the shared infrastructure); $mr = (r, R) \in \text{MR}_{i,k}$.
- Automated Verification Pipeline (AVP): $\text{AVP} : \text{Programs} \times \text{MR}_{\text{universe}} \times \mathbb{R}^+ \rightarrow \{\text{pass}, \text{fail}\}$; $\text{AVP}(s, mr, \epsilon_{\text{AVP}}^k)$ invokes the verification method per relation index k . Where the tolerance is fixed by the relation’s meta-pattern we abbreviate $\text{AVP}(s, mr) := \text{AVP}(s, mr, \epsilon_{\text{AVP}}^k)$.

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Mutation testing side.

- Mutation operator family: $MUT = \{\text{mut}_1, \dots, \text{mut}_5\}$, $j \in \{1, \dots, 5\}$ (open, extensible); $\text{mut}_1 = \text{mut}_C$, $\text{mut}_2 = \text{mut}_M$, $\text{mut}_3 = \text{mut}_G$, $\text{mut}_4 = \text{mut}_T$, $\text{mut}_5 = \text{mut}_F$.
- Signature: $\text{mut}_j : \text{Programs} \rightarrow 2^{\text{Programs}}$.
- Mutant: $s' \in \text{mut}_j(S_i)$.
- Alignment: $\text{align}(j) = j$.

Evaluation unit and pool decomposition (fixed). The evaluation unit is the (PUT, meta-pattern) cell (i, k) over the PUT's pooled admitted mutants $\text{mut}(S_i) = \bigcup_j \text{mut}_j(S_i)$; the operator index j is a per-mutant fiber label, not a cell axis.

$$\text{mut}(S_i) = \text{certified}_{i,k} \cup \text{killed}_{i,k} \cup \text{survive}_{i,k} \cup \text{unresolved}_{i,k},$$

with the four sets disjoint, mutually exclusive, and exhaustive.

- *equivalence-status evidence* (dual conditions): (E1) AVP-coherent: $\forall mr \in MR_{i,k} : \text{AVP}(S_i, mr) = \text{AVP}(s', mr)$; (E2) Output-equiv: $\forall x \in X_{K_{\text{eq}}} \sim \mathcal{D}_P : \|S_i(x) - s'(x)\| \leq \varepsilon_{\text{eq}}$. E1 $\vee$ E2 failure yields CONFIRMED_NON_EQUIVALENT; E1 $\wedge$ E2 agreement without a machine-checkable certificate yields EQUIVALENCE_UNRESOLVED; CERTIFIED_EQUIVALENT is reserved for certificate-bearing candidates (empty in the current pipeline).
- *killed determination*: $\text{killed}(s', MR_{i,k}) \Leftrightarrow \exists mr \in MR_{i,k} : \text{AVP}(S_i, mr) = \text{pass} \wedge \text{AVP}(s', mr) = \text{fail}$; killed mutants are confirmed non-equivalent (kill witness upgrade, main-text lemma).

Metric (fixed classical structure; strict and conservative readings).

$$\begin{aligned} \text{SMS}_{i,k} &:= \text{SMS}_{i,k}^{\text{strict}} = \frac{|\text{killed}_{i,k}|}{|\text{killed}_{i,k}| + |\text{survive}_{i,k}|}, \\ \text{SMS}_{i,k}^{\text{cons}} &= \frac{|\text{killed}_{i,k}|}{|\text{killed}_{i,k}| + |\text{survive}_{i,k}| + |\text{unresolved}_{i,k}|} \in [0, 1], \end{aligned}$$

each NA on an empty denominator. With $u_{i,k} = 0$ (the audited v4 campaign) both equal the classical ratio $|\text{killed}|/(|\text{mut}| - |\text{equiv}|)$.

LRCA engineering attribution layer (descriptive, not in SMS).

- Likely root-cause inventory: $C = \{C_1, \dots, C_5\}$ (open, extensible). C1 Likely semantic failure (residual label: no C2–C5 trigger); C2 Tolerance perturbation; C3 OOD; C4 Statistical-assumption violation; C5 Mutator artefact.
- LRCA output: each $s' \in \text{killed}$ annotated with $\text{root_cause}(s') \in C$.
- Descriptive quantities: $C1_share_{i,k}$, $\text{suspect_share}_{i,k}$; both NA on zero-kill cells.

Slicing and cross-class.

- Slicing: aligned $k = k^*(i)$ (the PUT's class-primary meta-pattern); cross $k \neq k^*(i)$.
- Cross-class: ΔSMS_c , $c \in \{A, B, C, D\}$; $\text{CV}(\Delta\text{SMS}) := \text{std}(\Delta\text{SMS}_c) / |\text{mean}(\Delta\text{SMS}_c)|$.

Theory-extension symbols (window, interval, and gap results).

- Observation: $\text{obs} : \mathcal{Y} \rightarrow \mathcal{Y}_{\text{obs}}$; observational equivalence $\equiv_{\text{obs}}$; computation map Φ_Q .
- Invariant family: $\Psi = \{\psi_1, \dots, \psi_5\}$ (non-redundant; open, e.g. ψ_6); edit operator edit with $P_{\text{edit}} = \text{edit}(P)$; effect map eff with fibers $\text{eff}^{-1}(\cdot)$.

- Counts (local to the theory subsections): n confirmed non-equivalent, κ killed, u unresolved, $u_{\text{neq}} \in [0, u]$.
- Coverage and gaps: $\text{Cov}(R)$; $\text{Gap}_{\text{aln}}(R)$; $\text{Gap}_{\text{str}}(R)$; exactness defect $\xi(R)$ (NA on zero kills; one-sided); kill signature $\text{sig}(m)$; universe M_{neq} , fibers M_j , weights w_j .
- Window quantities: violation magnitude $\varepsilon_{\text{viol}}(m)$; residual $\Delta_r = \sup_{x \in D_r} \varepsilon_r(x; P^*)$ over the applicable domain D_r ; margin $\mu_r = \varepsilon_{\text{tol}} - \Delta_r$; per-run noise bound $\bar{\eta}$ with execution multiplicity p ; repeats N , concentration constant c , output spread σ_{out} , deterministic noise η_{det} ; Lipschitz constant L_r ; crash threshold $\varepsilon_{\text{crash}}$; exception set $\mathcal{N}_{\text{exc}}$ (degeneration).
- Degenerate limit: $L = L_{\text{lim}} \wedge L_{\text{switch}} = L_{\text{equiv}} \wedge L_{\text{killed}} \wedge L_{\text{mut}}$.

Tolerance and sampling.

- Tolerance: $\varepsilon_{\text{AVP}}^k$ (relation-dependent), ε_{eq} (equivalence output tolerance).
- Sampling: $K_{\text{eq}} = 1000$ (E2 sample size), $N = 20$ (statistical replicates).

The notation skeleton (11 core concepts + three-state decomposition + SMS formula + AVP interface) is **fixed**; the content of MUT, $\mathcal{R}$, C, and cls is open to extension.

A.2 Root-Cause Decision Tree and Diagnostic Protocol

Five likely root causes (open):

Table 1. Root-cause decision tree and three-layer diagnostic protocol.

Code	Meaning
C1	Likely semantic failure (residual label: no C2–C5 trigger; the kill class SMS seeks)
C2	Numerical tolerance perturbation
C3	Out-of-distribution (OOD)
C4	Statistical-assumption violation
C5	Mutator artefact

Three-layer diagnosis with L0 artefact pre-scan:

- **L0 artefact pre-scan.** Double-blind review following the main manuscript’s mutant-pool prescreen and equivalence-detection protocol: dual-LLM cross-source review plus 20% manual sampling.
- **L1 tolerance robustness.** $N = 20$ replicates; fail ratio ≥ 0.80 considered stable.
- **L2 OOD triage.** For C/D classes, distinguish valid $\mathcal{D}_p^{\text{valid}}$ from OOD.
- **L3 statistical baseline.** For Wilcoxon/DTW, IID/stationarity pre-check on PUT samples.

Decision tree:

For each $s' \in \text{killed}_{i,k}$:

- **L1.** fail ratio $< 0.80 \Rightarrow \text{C2}$; otherwise $\Rightarrow \text{L2}$.
- **L2** (C/D classes). fail only in OOD $\Rightarrow \text{C3}$; otherwise $\Rightarrow \text{L3}$.
- **L3** (B/D classes + Wilcoxon/DTW). assumption violation $\Rightarrow \text{C4}$; otherwise $\Rightarrow$ artefact recheck.
- **Artefact recheck.** artefact evidence $\Rightarrow \text{C5}$; otherwise $\Rightarrow \text{C1}$.

Multi-label priority: $C5 > C4 > C3 > C2 > C1$.

LRCA threshold calibration (9-grid scan; lrca_calibration.json). Best `ood_band=0.02` with any `tolerance_multiplier` (3.0 / 10.0 / 30.0) gives 12 of 15 evaluable cells at `suspect_share` ≤ 0.20 . `tolerance_multiplier` has zero impact (L1 tolerance rarely triggered); `ood_band` is the only discriminator. Under the NA convention of the main text the pass-count denominator is the 15 evaluable cells, not 60; the main-text H4 discussion renders no pass/fail verdict because the pre-registered 60-cell mean is undefined on this data. The grid calibrates diagnostic labels only and is not an H4 hypothesis test.

A.3 Verification Protocol and Equivalence-Judgement Trade-Off

$$\text{AVP} : \text{Programs} \times \text{MR}_{\text{universe}} \times \mathbb{R}^+ \rightarrow \{\text{pass}, \text{fail}\}.$$

Per-relation verification: Cons. tolerance equality $|\text{LHS} - \text{RHS}| \leq \varepsilon$; Mono. / P – ord. Wilcoxon signed-rank $\alpha = 0.05$; Conv. convergence-order + asymptotic residual ratio; Traj. DTW distance threshold ε_{DTW} . AVP version = upstream infrastructure commit hash; our package embeds AVP source.

Three-candidate equivalence trade-off.

Table 2. Verification protocol and equivalence-judgement trade-off.

Determination	False-positive	False-negative	SMS bias
E1 alone	numerical coincidence + AVP-coherent	E1 false $\rightarrow$ non-equiv	Low (smaller denominator)
E2 alone	output match, MR differs (rare)	K_eq miss, full-space equal	High
E1 $\wedge$ E2	both mis-pass simultaneously	E1 $\vee$ E2 false $\rightarrow$ non-equiv	Slightly low (conservative: misclassified equivalents remain as unkillable survivors in the denominator)

E1 alone is fooled by insufficient AVP coverage; E2 alone by numerical coincidence on K_{eq} . E1 $\wedge$ E2 is the conservative complete instantiation; under L_{equiv} (with the switch component fixed) it reduces to classical bitwise equivalence outside the exception set $\mathcal{N}_{\text{exc}}$ (Lemma G.1). Counter-examples: E2 passes but E1 does not (rare; mutant coincides at K_{eq} but deviates at MR trigger points $\rightarrow$ non-equivalent); E1 passes but E2 does not (common; consistent within MR but numerical drift $> \varepsilon_{\text{eq}} \rightarrow$ non-equivalent).

B Experimental Subjects and Mutation-Operator Specialisations

B.1 Program Selection Rationale

(a) Library-stack coverage. numpy 2.4.4 (A1-B3 linear algebra / arrays), scipy 1.17.1 (A1 integrate, A2 linalg, B2 stats), scikit-learn 1.8.0 (C1-D3 surrogate / ML), Python scientific-computing’s de facto foundation stack (PyPI top-50, 2026-04). Uncovered: GPU / distributed (JAX, CuPy, dask), domain-specific (BioPython, Astropy, RDKit). Left to future work under the representativeness threat.

(b) Mathematical-structure coverage. ODE/PDE (A1, A3), direct linear algebra (A2), Bayesian analytic + MCMC + Monte Carlo (B1-B3), kernel + orthogonal polynomial + NN surrogate (C1-C3), convex optimisation + backprop + max-likelihood (D1-D3), covers 8 of 12 chapters in Numerical Recipes [2]. Uncovered: advanced PDE solvers (FEM, FV, spectral); FFT; interior-point/trust-region optimisation; symbolic/CAS. Left to future work.

(c) Comparison with benchmarks. DeepCrime (Humbatova et al. 2021): semantic class-D topical overlap on ML kernels. Defects4J (Just et al. 2014): no overlap, only mutation-as-fault-proxy reference. mutmut / cosmic-ray demos: general Python; the semantic operators extend the scientific-computing focus.

(d) Scale vs representativeness. Each PUT kernel file is 19-35 LOC (0.6-1.2 KB), wrapping its library calls behind the signature standardised to `program(x: float) → float`. Substantively smaller than industrial code (typically 1-10 KLOC).

Limitation. The `float → float` signature is a substantive constraint, not engineering trade-off. Industrial inputs are typically high-dimensional (CFD grids, MD particles, FEM matrices); scalarised PUTs upper-bound mutant semantic complexity and may systematically under-estimate SMS and cross-class differences on industrial PUTs (declared in the main manuscript's threats-to-validity section; future work validates on industrial-grade PUTs).

B.1.1 Systematic vs incidental. Syntactic tools may *occasionally* hit the semantic-mutation criteria in the main manuscript (a) (cross-function-boundary) or (c) (algorithm class), but occasionality undermines two engineering functions of semantic mutation. **(a) Deepening source-code understanding.** Designing `OS np.linalg.det(M) → np.sum(np.diag(M))` requires knowing the two are equivalent on diagonal matrices but not on general matrices (`det = ∏ eigvals` vs `sum(diag) = trace = ∑ eigvals`); syntactic tool AST traversal does not require this understanding even when similar replacements appear. **(b) Revealing deep faults.** Domain-semantic errors (physical-constant, unit conversion, boundary conditions, hyperparameter semantics, numerical-method order) are not AST-local; syntactic mutator design goals (operator typos, off-by-one, negation flips) have hit probability for domain errors much smaller than systematic semantic-mutator triggering, and lack repeatability. **Conclusion.** Syntactic tools occasionally producing mutants satisfying (a)(b)(c) is stochastic byproduct, not repeatable, not carrying engineering value. Systematic semantic mutation requires (a)(b)(c) to be design intent.

B.2 Per-class operator specialisations

Table 3. Per-class operator specialisations.

Operator	Representative specialisations across PUTs
mut_C Conservation-breaking	A1 Lorenz: add ϵ_{drift} to RHS (slow Hamiltonian drift); A2 LU: decomposition omits $k+1$ -th multiplier (breaks det conservation); B1 Beta-Bin: posterior omits normalisation; C1 GPR: covariance omits positive-definite diagonal term; D1 MLP: backprop omits one gradient term.

mut_M Monotonicity-breaking	A3 FDM: Δt coefficient occasionally negative; B2 MCMC: acceptance $\min(1, r) \rightarrow \min(0.95, r)$; C2 PCE: high-order coefficient sort inserts inversion; D2 SVM: decision-function sign flips near boundary.
mut_G Convergence-breaking	A1 Lorenz: RK4 $\rightarrow$ 1.5-order hybrid; A3 FDM: 2nd-order difference $\rightarrow$ 1st-order; B3 MC: doubling sample size does not $1/N$ -reduce variance; C3 NN-Surr: training-epoch truncation.
mut_T Trajectory-distorting	A1 Lorenz: state-vector y/z swap; B2 MCMC: insert independent-sampling segment; C3 NN-Surr: training-target slow phase shift; D1 MLP: hidden-layer periodic-mask activation.
mut_F Fidelity-order-breaking	A2 LU: partial pivoting degrades to no pivoting; C1 GPR: length-scale switches to coarse prior; C2 PCE: high-order term randomly retains low-order; D3 LR: regularisation occasionally large.

Per-class HP / OS / TF substitution rules (illustrative). **HP** on class a: tolerance / max_iter; on class c: GPR noise_level / length_scale; on class d: MLP hidden_dim / dropout. **OS** on class a: numerical-linalg API swaps ($\det \leftrightarrow \text{sum}(\text{diag})$); on class b: probability-distribution sampling swaps; on class c: surrogate-class swaps (GPR $\leftrightarrow$ RBF $\leftrightarrow$ NN). **TF** on class a: integration-order changes (RK4 $\rightarrow$ Euler); on class b: MC estimator changes.

Operator-level cross-table with cosmic-ray defaults. The 13 cosmic-ray default operators (the B.6 inventory) are AST-local: number_replacer (Num/Constant, CE partial, Δ literal values only); binary_operator_replacement (BinOp); comparison_operator_replacement (Compare); boolean_replacer (BoolOp); unary_operator_replacement (UnaryOp); keyword_replacer (NameConstant, CE keyword, Δ Bool literal); break_continue (Break/Continue); exception_replacer; remove_decorator; variable_inserter; variable_replacer; no_op; zero_iteration_for_loop (For). None correspond to OS / HP / TF / SI:

Table 4. Per-class operator specialisations.

Semantic class	Tool support	Coverage
OS API replacement	None	Δ 88.33% disjoint (incidental low-complexity hits)
HP	None	$\times$ tool inexpressible (0/72)
TF numerical-method order	None	$\times$ tool inexpressible (0/54)
SI / CF structural injection	None	$\times$ tool inexpressible (0/33)

Operator-level conclusion. All 12 default classes remain AST-local (BinOp / Compare / BoolOp / UnaryOp / NameConstant / Subscript / If / For / Break / Continue / decorator / except). Categorically, no entry recognises sklearn

/ scipy hyperparameter semantics (HP), numerical-method order (TF), or control-flow intent (SI / CF). For OS, low-complexity sub-expressions can be incidentally hit by BinOp; empirical 11.67% (the main manuscript’s AST-overlap audit) is consistent with 88.33% AST-disjointness lower bound.

B.3 Syntactic-Mutation Pilot Study

Before the 12-PUT generalisation in the main manuscript’s AST-overlap audit, a single-PUT lightweight empirical used scripts/run_cosmic_ray_a1.sh on a1 (file 934 B, AST-bound). Outcome: mutants_generated ~tens; $\geq 90\%$ belong to BinOp / Compare / 12 default classes; small killed-by-test_a1.py ratio (PUT unit tests are output-shape/type sanity checks; most cosmic-ray mutants not killed), an empirical manifestation of the semantic-mutation criteria in the main manuscript: syntactic-tool mutants are not aligned with MR-violation detection. The 12-PUT generalisation (the main manuscript’s AST-overlap audit) supersedes this pilot but corroborates the same conclusion at scale.

B.4 Expected Root-Cause Risk Profile

PUT-class $\times$ LRCA-layer risk weights. A numeric: C2 dominant ($\star\star\star$ on L1 tolerance). B probabilistic: C4 dominant ($\star\star\star$ on L3). C surrogate: C3 dominant ($\star\star\star$ on L2 OOD; $\star\star$ tolerance). D ML: C3 + C4 mixed ($\star\star\star$ L2; $\star\star$ L3).

Operator-PUT root-cause hotspots (expected). A: mut_C/G/F dominantly C2 (numerical tolerance), mut_M/T C1 (true semantic). B: dominantly C4 (statistical-assumption violations). C: mut_C/M dominantly C2/C1 on c1/c2, mut_T/F dominantly C3 on c1/c2; c3 (NN-Surr) row dominantly C5 (artefact). D: d1 row dominantly C5 (training-noise artefact); d2/d3 dominantly C1 / C3 mixed.

Expected suspect_share thresholds. A: 0.10-0.20 (acceptance ≤ 0.25); B: 0.20-0.35 (≤ 0.40); C: 0.20-0.30 (≤ 0.35); D: 0.25-0.40 (≤ 0.45). 60-cell average: ≤ 0.20 (= H4; acceptance ≤ 0.25).

B.5 Engineering-significance mapping

The aligned slice $k = k^*(i)$ is the H2 threshold-test slice; the cross slice $k \neq k^*(i)$ is control; 9 vacant cells (e, no MR exercised) are not formally adjudicated and are repurposed in the main manuscript’s R_sem/R_kill decoupling discussion as descriptive evidence for R_sem / R_kill decoupling. The v3 $\rightarrow$ v4 micro-change ($\Delta\delta = -0.009$, paired-role bootstrap 95% CI [-0.238, 0.207]) attributes to LLM-source diversity under a fixed prompt; this is the source-axis null contrast underpinning the main manuscript’s aligned-versus-cross results.

B.6 Default-Operator Overlap between Syntactic Mutation Tools

Reviewer-requested manual operator-class cross-reference between cosmic-ray’s 13 default operators (cosmic-ray 8.4.6, cosmic_ray.operators package) and mutmut’s 14 default operators (mutmut current main, src/mutmut/mutation/mutators.py). Both default-operator sets are first-order AST-local replacements; neither implements a cross-function-boundary, domain-knowledge-dependent, or algorithmic-class-changing mutation (the three necessary conditions in the semantic-mutation criteria of the main manuscript for semantic mutation).

Table 5. Mutmut vs cosmic-ray default-operator overlap.

Operator class	cosmic-ray default	mutmut default	Reaches the semantic-mutation criteria (a/b/c)?	Semantic-class reachability
Numeric literal ± 1	number_replacer	operator_number	(a) no, (b) no, (c) no	partial CE only (e.g. _RHO=28.0 $\rightarrow$ 27.5)
Binary arithmetic swap (+, -, $\times$, $\div$)	binary_operator_replacement	operator_swap_op (binary subset)	(a) no, (b) no, (c) no	partial OS only (incidental)

365	Comparison swap (<, ≤, >, ≥, ==, !=)	comparison_operator_replacement	operator_swap_op (compare subset)	(a) no, (b) no, (c) no	none
366	Boolean swap (and ↔ or)	boolean_replacer	operator_swap_op (bool subset)	(a) no, (b) no, (c) no	none
367	Unary remove / swap	unary_operator_replacement	operator_remove_unary_ops, operator_swap_op	(a) no, (b) no, (c) no	none
368	Keyword swap (True/False, etc.)	keyword_replacer	operator_keywords, operator_name	(a) no, (b) no, (c) no	none
369	break ↔ continue / return	break_continue	operator_keywords (subset)	(a) no, (b) no, (c) no	none
370	Exception class swap	exception_replacer	(none)	(a) no, (b) no, (c) no	none
371	Decorator removal	remove_decorator	(none)	(a) no, (b) no, (c) no	none
372	Variable-binding insert / replace	variable_inserter, variable_replacer	operator_arg_removal, operator_assignment	(a) no, (b) no, (c) no	none
373	no_op insertion	no_op	(none)	(a) no, (b) no, (c) no	none
374	Zero-iteration for loop	zero_iteration_for_loop	(none)	(a) no, (b) no, (c) no	none
375	String content tweak (XX prefix, case flip)	(none)	operator_string	(a) no, (b) no, (c) no	none
376	Lambda body → None/∅	(none)	operator_lambda	(a) no, (b) no, (c) no	none
377	Dict kwarg name (XX prefix)	(none)	operator_dict_arguments	(a) no, (b) no, (c) no	none
378	Sym./asym. string-method swap	(none)	string-method symmetric swap; string-method asymmetric swap	(a) no, (b) no, (c) no	none
379	Augmented → simple assignment	(none)	operator_augmented_assignment	(a) no, (b) no, (c) no	none
380	match case removal	(none)	operator_match	(a) no, (b) no, (c) no	none

Aggregate. Cosmic-ray $\cap$ mutmut: ~9 of 13 cosmic-ray operators have a near-equivalent in mutmut (numeric, binary, compare, boolean, unary, keyword, break-continue, variable, plus partial overlap on string-prefix vs XX); 4 cosmic-ray operators are unique (exception, decorator, no_op, zero-iteration-for); 6 mutmut operators are unique (string content, lambda, dict-kwarg, string-method swap, aug-assignment, match-case). The union is approximately 21 distinct first-order AST-local operator classes. None of the 21 can:

- (1) Cross a function-call or module-import boundary (e.g. replace `det(M)` with `sum(np.diag(M))` requires knowing the equivalence on diagonal matrices, outside any default-operator scope);
- (2) Depend on domain knowledge for legality (e.g. a hyperparameter swap that knows `noise_level=1e-4` is the load-bearing knob in a Gaussian Process Regression PUT requires PUT-class awareness);
- (3) Change the algorithmic class (e.g. swap `polynomial` → `spline` basis rewrites the surrogate’s mathematical structure).

The mutmut / cosmic-ray null intersection on the main manuscript’s AST-overlap audit’s HP/SI/TF classes (zero overlap) therefore reflects a **structural property of first-order AST-local mutation** rather than a tool-specific limitation, supporting the preventive-defence framing in the main manuscript.

C Experimental Procedure Details

C.1 Cross-Source Generation Protocol

Two-stage ablation.

Table 6. Cross-source generation protocol.

Version	Mutant pool	c-class primary MR	Use
v3	Single-source (Claude Opus 4.6)	P-ord. (pre-registered)	H2 baseline
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v4	Cross-source (Claude Opus 4.6 + GPT-5.4 + DeepSeek chat)	P-ord. (held at pre-registered)	Isolate LLM-source diversity
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Protocol (scripts/cross_source_campaign.py). (a) For each (PUT, operator), Claude / GPT / DeepSeek run $K = 3$ trials with identical prompt (the cross-source protocol in the main manuscript), temperature 0.7; source_tag propagated to filenames. (b) **V1-V4 mechanical validation** (src/p2/mutators/validation.py): V1 syntax (ast.parse), V2 executability, V3 non-triviality ($|y_{\text{mutant}} - y_{\text{original}}| > 1e-6$), V4 signature consistency. (c) **DeepSeek model**: deepseek-chat (113 tokens/call, 1.8 s), not deepseek-v4-pro (340 tokens/call, 11 s), quality-equivalent on a2_OS1 dry-run. (d) **Pool capacity**: 37 ops $\times$ 3 sources $\times$ 3 trials = 333 attempts; V1-V4 pass rate 89.5%; 298 confirmed mutants, of which 292 enter the v4 pool after final-stage rejection of duplicates and QA failures. Sources contribute nearly equally: Claude 101 / GPT 98 / DeepSeek 99 (Phase A engineering finding). (e) **Sampling** (scripts/build_pools.py, POOL_VERSION = v4): per-PUT 30 max, measured mean 24.3, range 10-30. c1 (GPR) has only 10 because c1_HP1 / c1_CE1 have V1-V4 near-zero pass (WhiteKernel noise_level $1e-4 \rightarrow 1e-1$ perturbation has minimal output impact, triggers V3 non-trivial failure, which supports the main manuscript’s R_sem/R_kill decoupling discussion).

Protocol asymmetry. v4 does not invoke reviewer LLM (cost / speed priority; a follow-up study will rerun dual-blind on the v4 grid, ~\$5-8 USD); v3 used the original Phase-1 dual-blind (Claude gen + GPT-5.4 review + DeepSeek arbitration). A fraction of the v3 $\rightarrow$ v4 source-axis shift may be slight v4 quality decline rather than LLM-source diversity.

C.1.1 Differential prompt protocol. The v3 $\rightarrow$ v4 micro-change of -0.009 (paired-role bootstrap 95% CI [-0.238, 0.207]) may reflect “three LLMs under identical prompt” rather than “true upper bound of LLM diversity”. Separation requires *one tailored differential prompt per LLM* (skeleton: scripts/run_differential_prompt.py).

Design (2 $\times$ 3 factorial, within-(PUT, operator)). Factor A: prompt template, 3 levels: **V_canonical** (control, identical to v4), **V_persona** (per-LLM identity: Claude “numerical-analysis editor”; GPT “scientific-software refactoring”; DeepSeek “library-API substitution specialist”), **V_cot** (Claude <thinking> tags; GPT “step by step”; DeepSeek stepped-reasoning). Factor B, LLM source (Claude / GPT-5.4 / DeepSeek chat). $K = 3$ per cell; total $37 \times 3 \times 3 \times 3 = 999$ trials.

Exit criteria. If $\max(\delta) - \min(\delta) < 0.05$: the v3 $\rightarrow$ v4 -0.009 source-axis null is robust. If ≥ 0.05 with prompt variant pushing delta past 0.474: H2 revised from “not met” to “prompt-conditional met”. If ≥ 0.05 but all delta < 0.474 : prompt sensitivity exists but H2 unchanged.

Resources. API ~\$18-30 USD; wall 3-4 h at concurrency 4; V1-V4 only, no reviewer LLM.

C.2 Manual Pilot and Initial Prescreen

The original cross-source protocol used 60% multi-LLM consensus (Claude Opus 4.6 + GPT-5.4 + DeepSeek chat unanimous) plus 40% manual injection by scientific-computing researchers. Each (mut_j, S_i) pair carried one prompt template containing PUT source, semantic intent of mut_j, prohibition against revealing specific MRs (avoids prompt leakage), and output requirements (syntactically correct, executable, single-point modification, diff < 10 lines). Reproducibility parameters: temperature 0.3 in manual-pilot stage (v4 cross-source uses 0.7 per §C.1); fixed seed; 5 candidates per prompt with 2-3 retained.

Double-blind review (Scheme C). Generator LLM-G (Claude Opus) and reviewer LLM-R (GPT-5.4) are cross-source with strict role separation. LLM-R sees only PUT original + mutant code, unaware of mut_j category / MR content / generator identity, outputting a triple (syntactic, executable, semantic-fault-injected). Double-confirmed mutants enter pool; inconsistencies enter manual arbitration ($\leq 10\%$). From the double-confirmed pool, 20% is sampled for manual review; manual-vs-LLM-R inconsistency $> 10\%$ triggers full manual downgrade. Same-source bias mitigation stratifies LLM-R by PUT class. Prompt-injection mitigation disables RAG / web search on the API path.

L0 pool prescreen. Each mutant passes static syntax (linters / type checkers), a unit self-test (simple inputs produce finite output), and double-blind sign-off. Target: 30-50 candidates per cell $\rightarrow$ 10-15 retained after prescreen.

C.3 Three-Layer Root-Cause Diagnosis

Decision tree, multi-label priority ($C5 > C4 > C3 > C2 > C1$), and output ($C1_share$, $suspect_share$) are stated in §A.2; the full pseudocode is reproduced there. LRCA does **not** modify SMS; killed set is not filtered. The main manuscript’s LRCA-mass results use the best calibrated combination ($ood_band = 0.02$, $tolerance_multiplier = 3.0$, $repeats = 20$); default-threshold results retained in `lrca_60cell_v3.json` as control. Interface with the preventive-defence risk profile in the main manuscript: L0 prescan $\leftrightarrow$ §C.2 dual-LLM double-blind; L1 tolerance $\leftrightarrow$ $N = 20$ repetition subprocess; L2 OOD $\leftrightarrow$ input-distribution definition; L3 statistical baseline $\leftrightarrow$ pre-check protocol.

C.3.1 Pilot calibration. A 2026 Q3 operator-level pilot ran 12 PUTs $\times$ 37 operators (36 primary PUT-class operator pairs plus one CF structural-injection specialisation; the conceptual semantic family count remains five; 12 `is_key=True` at $K=20$, 25 at $K=10$; 470 trials total). Pilot precursor quantities: R_sem ($V1-V4 \wedge operator_match$ pass rate), D_impl (median pairwise AST + literal + identifier Jaccard distance among confirmed mutants), R_kill (proportion of confirmed mutants killed by AVP under that PUT’s primary MR). Aggregate ($N=37$): R_sem median 0.50 mean 0.468 (0/37 with $R_sem=0$); D_impl median 0.42 mean 0.392 (1/37 with $D_impl \approx 0$); R_kill median 0.00 mean 0.189. All 37 operators produced ≥ 1 V1-V4 passing mutant, a necessary precondition for H1; the pre-registered H1 threshold itself (≥ 5 non-equivalent mutants on $\geq 9/12$ PUTs for ≥ 4 of 5 operator families) is nonetheless not met, as reported in the main text.

Key R_sem / R_kill decoupling pattern. $R_sem \geq 0.9 \wedge R_kill = 0$ combinations all on HP / TF / OS operators in c / d classes (e.g., `c1_CE1 noise 1e-4 $\rightarrow$ 1e-1` with R_sem 1.00, R_kill 0.00; `c3_HP1 relu $\rightarrow$ tanh`, `c3_TF1 max_iter 1000 $\rightarrow$ 5`, `d1_HP1 MLP  $\alpha$` , `d3_HP1 LR C`). $R_sem-high \wedge R_kill = 1$ combinations all on CE / OS operators in a / b classes (`a2_CE1 LU det`, `a2_OS1 prod $\rightarrow$ sum`, `b1_OS1  $\alpha/\beta$  swap`, `d1_TF1 label flip`). This is empirical evidence for R_sem / R_kill decoupling at cell level.

D Statistical Analysis Details

D.1 Reporting Conventions and Primary-Table Construction

An earlier draft defined a separate `SMS_unfiltered` variant “without LRCA filtering”. Under the final definitions LRCA never filters the killed set, so that variant coincides with SMS identically and has been removed; the diagnostic secondary view is the $C1$ -weighted SMS_{C1} of the main text, which is explicitly a product of two reported quantities rather than a second mutation score.

Primary-table construction.

Table 7. SMS variants and construction lemmas.

RQ	Primary statistics	Reporting format
RQ3	inst_rate, equiv_rate, C1_share, survive_rate	60-cell heatmap + 4-class marginals
RQ4	aligned-SMS vs cross-SMS; sparse ◦ vs dense ••equiv_rate	Cliff's delta + odds ratio + 95% bootstrap CI
RQ5	ΔSMS_c ($c \in \{A,B,C,D\}$); $\text{CV}(\Delta\text{SMS})$	sign test (df=3) + descriptive forest plot
RQ5 (desc.)	Spearman ρ + Kendall τ for SMS vs PC	scatter + dual-metric ranking comparison

Multiple comparisons. No cell-level family of p-values is claimed across the 60 cells (main text, statistical pipeline); the per-class Friedman tests carry a Bonferroni $\times 4$ correction. $N = 20$ bootstrap intervals: 1,000-iteration 95% CI ($B = 10,000$ for the headline H2 delta CI).

D.2 Effect-Size Sensitivity and Transformation Invariance

H4 cutoff sensitivity. Dense grid scan (cutoff $\in \{0.05, 0.10, \dots, 0.50\}$, step 0.05) on v4 data (scripts/h5_sensitivity.py, data/results/h5_sensitivity_v4.json), re-based on the 15 evaluable cells per the main text's NA convention (zero-kill cells carry no defined suspect_share):

Table 8. Cutoff sensitivity of the per-cell suspect-share pass count over evaluable cells.

cutoff	cells at or below cutoff	share of evaluable
0.05	12 / 15	80.0%
0.10	12 / 15	80.0%
0.15	12 / 15	80.0%
0.20 (paper)	12 / 15	80.0%
0.25	12 / 15	80.0%
0.30	12 / 15	80.0%
0.35	12 / 15	80.0%
0.40	12 / 15	80.0%
0.45	13 / 15	86.7%
0.50	13 / 15	86.7%

Conclusion. The per-cell pass count is flat across cutoffs $\in [0.05, 0.40]$: the evaluable cells' suspect_share distribution is strongly bimodal (twelve cells at or near 0, a small high-suspect tail carrying most of the pooled suspect mass), so the specific cutoff 0.20 is **not load-bearing**. An earlier draft reported this sweep over all 60 cells (12/60 flat) by coding zero-kill cells as suspect_share = 1; that coding is withdrawn with the H4 estimand discussion in the main text, and no pass/fail verdict attaches to this table.

Logit-transformation invariance. Cliff's delta is rank-based (function of $U/(n_1 \cdot n_2)$), mathematically invariant under any strictly monotone transformation (Romano 2006). Logit is strictly monotone on $(0, 1)$, so $\delta_{\text{logit}} \equiv \delta_{\text{raw}}$ is a construction result. **Not additional robustness evidence;** H2 robustness threats come from the main manuscript's 60-cell distribution results zero-mass dominance, not metric-scale choice.

Vacant-cell sensitivity. Excluding the 9 vacant cells (cells with no exercised MR) leaves the aligned-versus-cross verdict unchanged. In v4 the analysis becomes $n = 11$ aligned and $n = 40$ cross with $\delta = 0.323$ (PUT-cluster 95% CI [0.045, 0.599]). The exclusion does not move H2 across the pre-registered large-effect threshold. For the complete primary grid, 100,000 PUT-cluster resamples (seed 42) give $\delta = 0.314$, 95% CI [0.045, 0.594], with 0.00964 of resamples at or below zero (scripts/compute_rq2_cluster_bootstrap.py; data/results/rq2_cluster_bootstrap_v4.json).

Primary-MR withdrawal permutation-null. The auxiliary c-class check pools 15 SMS values, shuffles labels over the 3×5 (PUT, MR) grid, and applies the max-over-five withdrawal rule with 10,000 permutations (seed 42). The observed c-class mean is 0.314 versus null mean 0.349, one-sided $p = 0.989$ (data/results/c_class_permutation_v4.json). This supports treating the primary-MR convention as a protocol choice rather than as evidence of a hidden c-class inflation effect.

D.3 Power Analysis under Observed and Stipulated Effects

Plug-in bootstrap. With-replacement sample from observed aligned ($n=12$) and cross ($n=48$) v4 SMS pools; $N_{\text{sim}} = 5,000$, seed = 42 (scripts/compute_rq2_power.py, rq2_power_v4.json):

Table 9. Observed-distribution exceedance probabilities.

Threshold	Interpretation	Exceedance probability
delta > 0.000	Any-effect	0.978
delta > 0.147	Small	0.849
delta > 0.330	Medium	0.457
delta > 0.474	Large (H2)	0.155

Sample-size sweep ($n_{\text{aligned}} \in \{6, 12, \dots, 60\}$, $n_{\text{cross}} = 4 \times n_{\text{aligned}}$): the probability for delta > 0 is 0.909 at $n=6$, 0.978 at 12, and 0.994 at 18. These are observed-distribution exceedance probabilities, not power against a stipulated alternative. Observed $\delta_{v3} = 0.323 < 0.474$ and observed $\delta_{v4} = 0.314 < 0.474$; increasing sample only narrows CI.

Stipulated-alternative. Implemented in mixture-weight (scripts/compute_rq2_power_stipulated.py, $N_{\text{sim}} = 2,000$): SMS has heavy ties at zero ($45/60 = 0$), raw shifting is discontinuous (any $\varepsilon > 0$ jumps delta from 0.314 to 0.74). Mixture: aligned' = (probability w) sample from (observed_aligned + 0.001) + (probability $1-w$) sample from observed_aligned, calibrate w so $E[\text{delta}] \approx 0.474$. Calibrated $w = 0.373$, realised $E[\text{delta}] = 0.470$.

Table 10. Power analysis under observed and stipulated effects.

Stipulated truth	Criterion	Power
0.470 (target 0.474)	$\hat{\delta} \geq 0.474$ (point-estimate)	0.504
0.470 (target 0.474)	95% CI lower > 0 (any-effect)	0.895

At the nearest attainable mixture, $E[\delta] \approx 0.470$, the (12, 48) design returns $\hat{\delta} \geq 0.474$ in ~50% of replications. This supports the main manuscript’s aligned-versus-cross framing that “not met” is point-estimate, not effect-size.

D.4 Per-Class Friedman Tests

Per-class Friedman (3 SUTs $\times$ 5 MRs, v4 primary pool) with Bonferroni $\times$ 4: a $\chi^2=4.00$ raw 0.406 adj 1.000, $W=0.333$; b $\chi^2=10.34$ raw **0.035** adj **0.140**, $W=0.862$; c $\chi^2=5.60$ raw 0.231 adj 0.924, $W=0.467$; d $\chi^2=5.00$ raw 0.287 adj 1.000, $W=0.417$. After correction, no per-class result remains significant at family-wise $\alpha = 0.05$. Even b-class $W = 0.862$ (Cohen 1988 large concordance) is insufficient at $N = 3$ per class. **H3 verdict rests on the main manuscript’s cross-class consistency sign test**; per-class Friedman is sensitivity / descriptive only. Friedman main effect on the full 60 cells (v4: $\chi^2 = 16.76$, $p = 0.0022$; v3: 15.30, 0.0041) confirms MR rank differences but is logically independent from H3 cross-class consistency.

Mixed-effects unavailability. Primary $\text{sms} \sim C(\text{class}) + C(\text{operator}) + C(\text{class}):C(\text{operator}) + (1 \mid \text{put})$ returned Singular (insufficient column rank for class $\times$ operator interaction; $N=60$ too small for 11-d fixed-effects + 12 PUT random intercepts). Fallback $\text{sms} \sim C(\text{class}) + C(\text{operator}) + (1 \mid \text{put})$ had PUT random-intercept variance hit boundary 0 (degenerates to OLS). Fallback class p-values (descriptive only): b vs a 0.275 / c vs a 0.892 / d vs a 0.991. RQ5 conclusion shifts to four-piece presentation (class means + sign test + Friedman + forest plot).

D.5 Pattern-coverage operationalisation

Per SUT, (MR_k, R_outcome $\in \{\text{True}, \text{False}\}$) binary-tuple coverage: 5 MRs $\times$ 2 outcomes = 10 cells, PC = #triggered / 10. Simplest baseline (per-SUT granularity, not distinguishing mutants). PC range over 12 SUTs (v4): [0.500, 1.000], mean 0.750 (rq4_pattern_coverage_v4.json). Pairing per PUT: Spearman $\rho = 0.163$; Kendall $\tau = 0.136$ (descriptive).

Power caveat. At $n = 12$, Spearman 95% CI $\approx [-0.5, +0.6]$ (Fisher-z approximation); cannot distinguish “zero correlation” from “moderate positive” or “moderate negative”. The descriptive reading is “no correlation detected at $n = 12$ ”, not “correlation does not exist”.

Within-class descriptive. In the b class, b2 (PC = 1.0) has mean SMS 0.133 while b1 (PC = 0.7) has 0.20: higher PC, lower SMS. In the c class the ordering reverses: c3 (PC = 1.0) has mean SMS 0.148, c1 (PC = 0.9) has 0.12, and c2 (PC = 0.8) has 0.0. The within-class PC-SMS ordering is therefore not stable across classes, consistent with the headline finding of no detectable correlation at $n = 12$; an earlier draft reported v3 values here and read the b-class pattern as uniformly “higher PC, lower SMS”, which the v4 numbers do not support. A confirmatory study would need $n \geq 30$ and a PC definition that incorporates the mutant dimension.

E Stakeholder Deployment Considerations

E.1 Air-Gap Deployment Constraints

The stakeholder-analysis workflow in the main manuscript depends on external LLM API calls (Claude / GPT / DeepSeek), incompatible with most regulated air-gapped V&V workflows:

Table 11. Air-gap deployment constraints.

Domain	Standard	Air-gap requirement
Nuclear	IEC 60880	Air-gapped build for safety-critical software
Aerospace	DO-178C	Tool-qualified offline build for DAL-A/B; LLM API outside qualified-tool envelope
Medical	IEC 62304	Class C software lifecycle requires offline reproducible build
Automotive	ISO 26262	ASIL-D requires offline tool chain; cloud LLM non-compliant

Mitigation paths. (a) Self-hosted open-weight LLMs (Llama / DeepSeek local inference), capability gap vs API-served frontier models; quantitative impact on SMS untested. (b) Offline-cached pre-generated mutant pools with commit-hash + raw-response signature locking, adopters reproduce a published pool offline (this paper’s package supports this), but generating new pools requires external LLM access. SMS for air-gapped industrial deployment is a future research direction.

E.2 Quarterly batch audit workflow

The original per-PR GitHub Actions YAML was removed because its PR-CI threshold 0.10 contradicted the Appendix E.3 audit threshold 0.20 and propagated stakeholder-facing CI assumptions into adopters’ pipelines.

Replacement: quarterly batch audit:

Table 12. Quarterly batch audit workflow.

Step	Description	Cost
1	Offline mutant-pool generation (per PUT $24\text{-}30 \times 12$ PUTs ≈ 300)	LLM API \$5-15; ~30 min
2	<code>sms_campaign.py --track 2</code>	4-core laptop ~10-20 min
3	<code>run_lrca.py</code>	< 1 min
4	MR-design team review of MRs with aligned-cell SMS significantly below the historical aligned mean (0.213)	1-2 h meeting

Quarterly: API ~\$10 + automation 30 min + manual 1-2 h ≈ 0.5 person-day. Affordable. Per-PR gating not recommended (LLM latency 5-30 s + cost + non-determinism + air-gap incompatibility); adopters needing per-PR should use coverage / unit-test gates. SMS does not replace domain-expert MR physical-reasonableness judgement nor product-requirements review.

E.3 Verification-and-Validation Documentation

This appendix subsection makes **no normative claim** toward NRC, FDA, or ISO 26262 review teams. Within single-output kernels scope, no traceable mapping exists to current IEC 60880 / ISO 26262 / DO-178C / ASME V&V 20-2009 normative bodies.

V&V documentation for scientific computing software (per ASME V&V 20-2009 §3 and similar guides) requires quantifiable test-adequacy evidence; current documentation often relies on code coverage + MR lists + SME signatures. SMS may appear as **research-grade supplementary evidence** alongside coverage and MR lists: (a) aligned-cell SMS per critical PUT; (b) LRCA three-layer diagnosis (C1 readout / C2-C5 attribution); (c) 60-cell matrix visualisation. Reviewers can independently run REPRODUCIBILITY.md.

Original threshold recommendations (aligned-cell SMS ≥ 0.20 / $0.30 + C1_share \leq 0.20$) were removed: no normative backing in IEC / ISO / ASME; misreads as enforce-ready. **SMS reported as descriptive supplementary evidence**, with adequacy judgements left to V&V reviewers case-by-case.

Conceptually complementary to ASME V&V 20-2009 §3 code verification (numerical-solver correctness) vs SMS (MR-set fault-detection adequacy). Substantial scale gap to multi-module CFD codes; incorporation into V&V standards body would require large-scale empirics + multi-year dialogue with ASME V&V 20 / IEEE 1012 / IEC 60880 committees. **No advocacy that SMS enter any normative certification system in 2027.** SMS does not replace SME signatures, FMECA, PIRT, system-level V&V, or ASME V&V 20 §3 numerical-solver verification, it is one link in the evidence chain, not a single-point qualification.

F Threats to Validity: Detailed Mitigation

F.1 Internal threats

Hoeffding-style false-equivalence bound (referenced by main-text Limitation 1). For a mutant whose true per-sample disagreement probability against the original program under $\mathcal{D}_P$ is at least p_0 (at the E2 tolerance), the probability that all K_{eq} independent E2 samples agree, producing a false equivalence verdict, is at most $(1 - p_0)^{K_{eq}} \leq \exp(-K_{eq} p_0)$. At $K_{eq} = 1000$: $p_0 = 0.005$ gives $\leq e^{-5} \approx 0.7\%$; $p_0 = 0.01$ gives $\leq e^{-10} \approx 4.5 \times 10^{-5}$. The bound conditions on p_0 and says nothing about mutants whose disagreement is concentrated below the E2 tolerance; those are governed by the E1 arm and remain the residual risk the K_{eq} sweep (deferred) would quantify.

Table 13. Internal threats.

Threat area	Threat	Mitigation	Residual
LLM generation	LLM generation reproducibility	§C.2 reproducibility parameters in package; dual-LLM cross-source + 20% manual; reproducibility rests on archived raw responses and frozen mutant pools (no user-facing seed)	LLM training data may be updated post-submission

Equivalence approximation	Probabilistic equiv (K_eq=1000)	the main manuscript's equivalence-judgement section declares probabilistic approximation; Hoeffding-style false-equiv bound	K_eq sweep $\in \{500, 1000, 2000\}$ deferred to future work
Shared infrastructure	Shared-infrastructure dependency	AVP version pinned to the upstream commit hash; package embeds AVP source; the arXiv infrastructure technical report is disclosed in the main text	None within scope
Root-cause diagnosis	LRCA multi-label boundary	§A.2 priority C5 > C4 > C3 > C2 > C1; multi-label co-occurrence table in package; root_cause is “likely root cause” not definitive	C2-C5 may multi-trigger on B/D classes
Pool size	Mutant-pool size	Pool expanded to mean 17.4: delta moved 0.321 $\rightarrow$ 0.323, CI narrowed; effect size is intrinsic ceiling, not pool dilution	Some PUTs < 30 (cache-bound)
LLM non-determinism	LLM non-determinism	Multi-turn de-dup; K=10/20 repetitions; raw prompt+response committed for direct reuse	Claude Opus subscription lacks seed control
Protocol asymmetry	Protocol-implementation gap v3 vs v4	v4 used 3 providers (Claude 101 / GPT 98 / DeepSeek 99) without dual-blind; v4 LRCA-evaluable cells increase 12 $\rightarrow$ 15 and macro C1 share changes 0.821 $\rightarrow$ 0.837	Complete separation requires a follow-up dual-blind rerun on the v4 grid

The protocol-asymmetry threat (dual-blind vs V1-V4 only) applies to the v3 $\rightarrow$ v4 source-axis contrast ($\Delta\delta = -0.009$, paired-role bootstrap 95% CI [-0.238, 0.207]).

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F.2 External / construct / conclusion threats

Table 14. External / construct / conclusion threats.

Threat area	Threat	Class	Mitigation
Representativeness	12-PUT representativeness	External	follows the shared site selection; open class principle (cls extensible); scaling to MD/QC/CFD
Statistical power	Cross-class statistical power	Conclusion	H3 framed exploratory; mixed-effects unavailable (Singular at N=60); shifted to class means + sign test + Friedman + forest plot
LLM homogeneity	LLM homogeneity bias	Construct	§C.2 three-LLM rotation + PUT-class subdivision + 20% manual sampling; third-party LLM family validation deferred
LLM-source shift	LLM-source distributional shift	External	Hit distribution DeepSeek 11/15, Claude 4/15, GPT 0/15; the main manuscript's AST-overlap audit argument is categorical AST-locality, not hit frequency; HP/SI/TF zero-overlap across all 3 sources unaffected
Higher-order mutation	HOM equivalence	External	Tool-unreachability claim confined to first-order syntactic tools; HOM empirical testing listed as future work

Conclusion threats: multiple comparisons handled per the main-text policy (per-class Bonferroni $\times 4$; no cell-level family claimed; §D.1); N=20 stability handled by 1,000-iteration bootstrap CI (§D.3). Construct threat, SMS measures epistemological semantic detection, not production engineering value (out of scope here).

G Degeneration to Classical Mutation Score: Full Proof

G.1 Notation cross-reference

PUT S_i , mutation operator family mut (syntactic or semantic), pooled mutant set $\text{mut}(S_i)$, MR-label set, equivalence tolerance ε_{eq} , equivalence sample size K_{eq} , AVP tolerance ε_{AVP} . Pool decomposition (three-state refinement, §A.1): $\text{mut}(S) = \text{certified} \cup \text{killed} \cup \text{survive} \cup \text{unresolved}$ (disjoint). Strict SMS formula (the reported reading; in the $u = 0$ regime it equals the classical form):

$$\text{SMS}_{i,k} = \frac{|\text{killed}_{i,k}|}{|\text{killed}_{i,k}| + |\text{survive}_{i,k}|} = \frac{|\text{killed}_{i,k}|}{|\text{mut}(S_i)| - |\text{equiv}_{i,k}|} \quad (u_{i,k} = 0)$$

G.2 Degenerate-limit definition

The degenerate limit $L = L_{\text{equiv}} \wedge L_{\text{killed}} \wedge L_{\text{mut}}$ consists of three **joint conditions** (paired axes), each controlling one layer of the SMS formula. L1-L6 are not 6 independent axes; the pairing reflects dependency structure across the SMS formula layers. Equivalently, the same conjunction regroups orthogonally as $L = L_{\text{lim}} \wedge L_{\text{switch}}$, where $L_{\text{lim}} = L1 \wedge L2 \wedge L3$ collects the quantitative limit axes and $L_{\text{switch}} = L4 \wedge L5 \wedge L6$ collects the discrete configuration switches (reference-anchored identity relation; rule-based syntactic operators; imperative deterministic PUT class).

Table 15. Degenerate-limit definition.

Joint condition	Component axes	Pairing rationale
L_{equiv} (Lemma G.1)	L1: $\varepsilon_{\text{eq}} \rightarrow 0$; L2: $K_{\text{eq}} \rightarrow \infty$	L1 alone: equiv stays probabilistic (K_{eq} cannot cover $\mathcal{D}_P$). L2 alone: bitwise equality diluted by ε_{eq} . Both simultaneously degenerate equiv to classical behavioural equivalence outside the exception set $\mathcal{N}_{\text{exc}}$.
L_{killed} (Lemma G.2)	L3: $\varepsilon_{\text{AVP}}^k \rightarrow 0$; L4: MR-label set = $\{\text{MR}_{\text{eq}}^{S_i}\}$, the reference-anchored identity relation ($r = \text{id}$, $R(y, y') \equiv y = y'$, follow-up output compared against the recorded reference output of S_i)	L3 alone: $\varepsilon_{\text{AVP}} \rightarrow 0$ still allows non-trivial MR (monotonicity, convergence). L4 alone: equality still carries ε_{AVP} tolerance; without the identity restriction the comparison $S_i(x)$ vs $s'(r(x))$ retains a metamorphic follow-up and is not the classical same-input oracle. Both simultaneously degenerate killed to classical difference detection.

L_{mut} (Lemma G.3)	L5: $\text{mut}_j \rightarrow$ rule-based syntactic (Mothra AOR/ROR/SDL/CRP); L6: $\text{cls}(I) \subseteq \{\text{imperative deterministic}\}$	L5 alone: syntactic ops on probabilistic/ML may still trigger domain-semantic subsets. L6 alone: imperative programs still mutable by OS/HP/TF/SI. Both simultaneously degenerate $\text{mut}(S)$ to Jia & Harman syntactic mutant set.
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G.3 Lemmas for the Pool Decomposition

Lemma G.1 (equiv degeneration; exception-set qualification). Assume $\text{supp}(\mathcal{D}_P) \supseteq \mathcal{X}_{\text{adm}}$ and fix the switch component L_{switch} . Under L_{equiv} ($L1 \wedge L2$), semantic-class equivalence ($E1 \wedge E2$) degenerates to classical behavioural equivalence **outside an exception set** $\mathcal{N}_{\text{exc}}$: over floating-point (finite) input domains, $\mathcal{N}_{\text{exc}}$ is the finite set of pathological inputs (Not-a-Number propagation and overflow/underflow traps), empty in the exact discrete case; under the continuum idealisation, $\mathcal{N}_{\text{exc}}$ has $\mathcal{D}_P$ -measure zero.

Proof. Under the fixed L_{switch} the relation set is the reference-anchored identity relation $\text{MR}_{\text{eq}}^{S_i}$, whose AVP check of a program Q compares Q 's outputs against the recorded reference outputs of S_i on the executed inputs. The E2 limit below yields exact output equality on its covered domain. Exact equality makes the original and mutant AVP verdicts identical for every non-negative AVP tolerance, so E1 is automatically satisfied on the E2 limit. No $\varepsilon_{\text{AVP}} \rightarrow 0$ premise is used here; that L3 axis belongs only to the killed-set degeneration of Lemma G.2. E2 ($\|S_i(x) - s'(x)\| < \varepsilon_{\text{eq}}$ for K_{eq} samples) degenerates in two regimes. Over floating-point (finite) input domains, L2 ($K_{\text{eq}} \rightarrow \infty$) with full-support sampling ($\text{supp}(\mathcal{D}_P) \supseteq \mathcal{X}_{\text{adm}}$) covers every admissible input almost surely, and L1 ($\varepsilon_{\text{eq}} \rightarrow 0$) sharpens the per-input test to exact equality, giving $\forall x \in \mathcal{X}_{\text{adm}} \setminus \mathcal{N}_{\text{exc}} : S_i(x) = s'(x)$ with $\mathcal{N}_{\text{exc}}$ the finite pathological set ($\mathcal{N}_{\text{exc}} = \emptyset$ in the exact discrete case). Under the continuum idealisation, sampled agreement at $\varepsilon_{\text{eq}} \rightarrow 0$ over an almost-surely dense full-support sample forces equality $\mathcal{D}_P$ -almost everywhere, i.e. outside a $\mathcal{D}_P$ -measure-zero $\mathcal{N}_{\text{exc}}$; pointwise equality on all of $\mathcal{X}_{\text{adm}}$ is not claimed in this regime, and the degenerate equivalence claim is read on $\mathcal{X}_{\text{adm}} \setminus \mathcal{N}_{\text{exc}}$ throughout. This matches Jia and Harman [1] §3 classical equivalent-mutant definition up to the exception set.

Support-assumption counterexample (non-removability). Let $\mathcal{X}_{\text{adm}} = [0, 2]$ but $\text{supp}(\mathcal{D}_P) = [0, 1]$, and let s' differ from S_i only on $(1, 2]$. Every E2 sample agrees at every K_{eq} , so the limit judgement certifies equivalence, yet s' is behaviourally non-equivalent on $\mathcal{X}_{\text{adm}}$; the degeneration target is therefore false without $\text{supp}(\mathcal{D}_P) \supseteq \mathcal{X}_{\text{adm}}$.

Lemma G.2 (killed degeneration). Under $L3 \wedge L4$, killed degenerates to classical same-input difference detection.

Proof. L4 restricts the MR-label set to $\{\text{MR}_{\text{eq}}^{S_i}\}$ with identity source transformation ($r = \text{id}$), $R(y, y') \equiv y = y'$, and the follow-up output compared against the recorded reference output of S_i . The killed predicate requires $\text{AVP}(S_i, mr) = \text{pass}$, which holds identically for the reference-anchored relation, and $\text{AVP}(s', mr) = \text{fail}$, which under L3 ($\varepsilon_{\text{AVP}} \rightarrow 0$) is exactly $\exists x : s'(x) \neq S_i(x)$: classical same-input difference detection against the original program. Without the identity restriction the comparison $S_i(x)$ vs $s'(r(x))$ retains a metamorphic follow-up and is not the classical oracle, and without the reference anchor an identity relation evaluated within a single deterministic program passes trivially on both S_i and s' and detects nothing; both clauses of L4 are therefore load-bearing.

Lemma G.3 (mut degeneration). Under $L5 \wedge L6$, the pooled $\text{mut}(S_i)$ degenerates to the syntactic mutant set of Jia and Harman [1].

Proof. L5 switches every generator family to rule-based syntactic operators (AOR, ROR, SDL, CRP, UOI, standard Mothra/Proteum sets); L6 restricts PUTs to imperative deterministic programs, excluding triggering conditions for semantic operators on probabilistic/ML programs. Under this configuration, $\text{mut}(S_i)$ is the literature-defined syntactic mutant set, independent of domain semantics.

G.4 Degeneration Theorem: Detailed Proof

Theorem G (SMS $\rightarrow$ MS degeneration; stated in the main text as the degeneration theorem). Fix the switch component $L_{\text{switch}} = L4 \wedge L5 \wedge L6$ and assume $\text{supp}(\mathcal{D}_P) \supseteq \mathcal{X}_{\text{adm}}$. Along the limit component $L_{\text{lim}} = L1 \wedge L2 \wedge L3$, outside the exception set $\mathcal{N}_{\text{exc}}$ of Lemma G.1,

$$\text{SMS}_{i,k} \xrightarrow{L_{\text{lim}}} \text{MS}_i := \frac{|\text{killed}_i^{\text{classic}}|}{|\text{mut}^{\text{syntax}}(S_i)| - |\text{equiv}_i^{\text{classic}}|}$$

Proof. Combine Lemmas G.1-G.3:

- Numerator: under $L3 \wedge L4$, $\text{killed}_{i,k} \rightarrow \text{killed}_i^{\text{classic}}$ (Lemma G.2); L4 makes $\text{MR}_{i,k}$ trivial over k (only the reference-anchored identity relation remains), so the subscript k degenerates.
- Denominator $|\text{mut}(S_i)|$: under $L5 \wedge L6 \rightarrow |\text{mut}^{\text{syntax}}(S_i)|$ (Lemma G.3).
- Denominator $|\text{equiv}_{i,k}|$: under $L1 \wedge L2$ with the support assumption $\rightarrow |\text{equiv}_i^{\text{classic}}|$ (Lemma G.1; outside $\mathcal{N}_{\text{exc}}$), and the unresolved set empties into the classical dichotomy, so the strict and conservative readings coincide in the limit.

Substituting into the SMS formula gives MS_i ; the restriction to any single generator family is immediate.

Corollary G (LRCA trivialisation). Under L , $C = \{C1, \dots, C5\}$ degenerates to $\{C1\}$.

Sketch. Each of C2-C5's triggering precondition depends on at least one L_j being violated. When the three joint conditions $L_{\text{equiv}} \wedge L_{\text{killed}} \wedge L_{\text{mut}}$ hold simultaneously, every non-trivial-space dimension (MR non-triviality, AVP tolerance non-zero, non-empty equiv set, MR-design DOF, class-mapping openness) closes; C2-C5 triggering set becomes empty (read off from §A.2 decision tree). The per-C_k to per-L_j minimum-sufficient mapping depends on the main manuscript's LRCA-diagnosis engineering thresholds; we do not claim one-to-one correspondence at formal level, readers can trace through §A.2 + §C.3. Under L , $\text{suspect_share} \rightarrow 0$, LRCA reports only C1, SMS degenerates to single-layer metric consistent with Jia and Harman [1] MS engineering structure.

Empirical consistency. Theorem G and the LRCA-trivialisation corollary jointly show that the SMS-based empirical conclusions (Cliff's delta in the aligned-versus-cross results, Friedman χ^2 in the cross-class consistency results, and Spearman rho in the SMS-vs-Pattern-Coverage analysis) are structurally consistent with existing Jia and Harman [1] literature in classical syntactic-mutation scenarios, no metric-level semantic fragmentation.

G.5 Independent formal audit of the theory extension

The theory-extension results (the detection-window, interval-soundness, and gap-attribution theorems with their lemmas, corollary, and remarks, and the repaired degeneration theorem above) were checked by an independent formal audit against a fixed eight-item checklist: (1) every theorem premise is defined in the body; (2) no circularity; (3) the scope of the AVP-determinism assumption of the kill-witness lemma; (4) consistency of the gap theorem's S5/exact-checker premises with the ξ reporting rule; (5) satisfiability of the window theorem's constants and Lipschitz hypothesis per PUT class; (6) the degeneration theorem's support assumption and exception-set wording; (7) notation conformance

against the frozen symbol registry; (8) line-by-line checkability of the proofs. The audit raised blocking and minor findings (premise hygiene, the ξ empty-kill convention and one-sidedness, the reference-anchored degenerate oracle, estimand definedness); all were repaired and verified closed in follow-up rounds, for a final verdict of PASS. The audit protocol and the signed report are archived with the replication materials.

G.6 Interval soundness: full proofs

Proof of the protocol-level kill-witness lemma. A kill supplies an mr for which $AVP(S_i, mr) = \text{pass}$ and $AVP(s', mr) = \text{fail}$. Hence E1 fails on that relation. The three-state protocol defines E1 or E2 failure as CONFIRMED_NON_EQUIVALENT; no determinism or tolerance-margin premise is needed. Therefore no killed mutant is unresolved. $\square$

Scope of the determinism hypothesis (quantitative output witness). The evaluation procedure of a relation fixes its executed input tuple (seeded input generation is part of the procedure), runs the program on that tuple, applies the observation obs , and computes the verdict by a fixed threshold functional at tolerance ε_{AVP} . For deterministic PUTs the verdict is a fixed function V_{mr} of the tuple of observed outputs on the common executed inputs. For stochastic PUTs the pipeline replaces each AVP call by a majority vote over $N = 20$ repetitions; the verdict is then a fixed function of the N -repeat aggregated observation, and every occurrence of “observed output” is read at that aggregated level: the witness produced below is a divergence of aggregated observations, in the AVP verdict semantics, not a claim about any single random run. (The LRCA layer’s 0.80 fail-ratio cutoff is a separate robustness annotation on killed mutants and plays no role in the kill verdict.) The stability hypothesis (clause R2, §G.8) is not removable: with relation residuals $\tau - \delta_1$ (pass on S_i) and $\tau + \delta_2$ (fail on s'), $\delta_1 + \delta_2$ arbitrarily small, a kill can arise from pointwise output divergence below ε_{eq} , and bare determinism cannot yield the required witness.

Proof of the quantitative output-witness proposition. Fix mr with $AVP(S_i, mr) = \text{pass}$ and $AVP(s', mr) = \text{fail}$. Suppose every executed input x satisfied $\|\text{obs}(\Phi_{S_i}(x)) - \text{obs}(\Phi_{s'}(x))\| \leq \varepsilon_{eq}$. Then the observed-output tuple of s' is a pointwise $\leq \varepsilon_{eq}$ perturbation of that of S_i , and R2 forces V_{mr} to return the same verdict on both, contradicting pass/fail. Hence some executed x has divergence exceeding ε_{eq} : the divergence witness of the three-state classification. $\square$

Proof of the interval theorem. By the kill-witness lemma the killed count κ lies inside the confirmed non-equivalent population n , so the ground-truth denominator is $n + u_{\text{neq}}$ with the same numerator κ . (*Interval and attainability.*) For fixed $\kappa \geq 0$, $n \geq 1$ the map $t \mapsto \kappa/(n + t)$ is non-increasing on $t \geq 0$; applying it to $0 \leq u_{\text{neq}} \leq u$ gives the sandwich, with both endpoints attained at $u_{\text{neq}} = u$ and $u_{\text{neq}} = 0$, each consistent with the available evidence. (*Width.*) $\kappa/n - \kappa/(n + u) = \kappa u/(n(n + u)) = \text{SMS}^{\text{strict}} \cdot u/(n + u)$. (*Monotonicity, four motions.*) (1) An equivalence certificate on an unresolved survivor maps $(n, \kappa, u) \rightarrow (n, \kappa, u - 1)$: the upper endpoint is unchanged and the lower endpoint rises, so the interval weakly narrows (strictly when $\kappa > 0$, $u \geq 1$). (2) A divergence witness on an unresolved survivor maps $(n, \kappa, u) \rightarrow (n + 1, \kappa, u - 1)$: the lower endpoint $\kappa/(n + u)$ is unchanged and the upper endpoint falls to $\kappa/(n + 1)$, so the interval weakly narrows; unresolved mutants are survivors by the kill-witness lemma, so κ is unchanged. (3) An unresolved survivor killed by a newly added MR acquires the witness, $(n, \kappa, u) \rightarrow (n + 1, \kappa + 1, u - 1)$, an instance of motion (4) with $\Delta = 0$, $j = 1$. (4) For $R \subseteq R'$ with the three-state classification held fixed, OR-aggregation over a superset can only add kills: with $\Delta \geq 0$ new kills among confirmed survivors and $j \geq 0$ among unresolved survivors, the counts update to $n' = n + j$, $\kappa' = \kappa + \Delta + j$, $u' = u - j$, and $\text{SMS}^{\text{strict}'} = (\kappa + \Delta + j)/(n + j) \geq (\kappa + j)/(n + j) \geq \kappa/n$ (since $\kappa \leq n$), while $\text{SMS}^{\text{cons}'} = (\kappa + \Delta + j)/(n + u) \geq \kappa/(n + u)$: both endpoints are non-decreasing. The proviso is not removable: if the classification were recomputed against R' , the E1 quantification domain would grow and a new relation with differing verdicts in the non-kill direction would convert an unresolved survivor into a confirmed one, a

motion-(2) event lowering the strict endpoint. Across all four motions the conservative endpoint never decreases: it is a monotone lower bound of the ground-truth score under every admissible evidence update. $\square$

G.7 Block structure: full proofs

Proof of the closure lemma (exact checkers are strong MRs). The violation-set identity is the exact-checker definition read as a set equation. Each stratum invariant is a predicate over the observable behaviour, and the tolerance-indexed satisfaction evaluates that predicate on the observed behaviour; membership of the violation set therefore depends on a program only through its observed behaviour, so observationally equivalent programs violate together: the set is a union of equivalence classes, i.e. closed, and the strong-MR definition applies. Properness: the union of two distinct strata's violation classes is closed but pins to neither stratum. $\square$

Premise reading. (i) reads: every admitted non-equivalent mutant is stratified, violating its unique target stratum (S3 witness) and satisfying all others (S5), so the fibers $\{M_j\}$ partition M_{neq} and $\sum_j w_j = 1$; mutants outside this reading (active-off-taxonomy, multi-stratum violators) are excluded by the premise, and their off-diagonal kills are exactly what ξ registers (the diagnostic is one-sided: premise violations that kill only on the labelled stratum leave ξ unchanged, so observed ξ lower-bounds the deviation mass). (ii) attaches to every $r \in R$ a stratum label, unique by the non-redundancy convention. (iii) imports, for every $r \in R$, the residual budget and margin-dominance clause R1 of §G.8: the S5-satisfied stratum contributes zero violation, the budget bounds the executed residual by $\Delta_r + p\bar{\eta}$, and R1 places that below ϵ_{tol} .

Proof of the gap theorem. (No cross-stratum flag.) Let $r \in R$ be an exact checker of ψ_l and $m \in M_j$ with $l \neq j$. By (i), m satisfies ψ_l ; by (iii) the executed residual of r on m stays below ϵ_{tol} , so the tolerance verdict is pass and r does not flag m . The kill predicate requires a fail on the mutant, so r does not kill it. (*Uncovered fibers survive.*) If $j \notin \text{Cov}(R)$, every $r \in R$ has label $l \neq j$, so no r flags any member of M_j and OR-aggregation yields no kill. (*Block diagonality.*) Indexing rows by fibers and columns by checker strata, the argument shows every off-diagonal entry is zero. (*Decomposition.*) The fibers partition M_{neq} , so $\text{SMS}(R) = \sum_j w_j \text{SMS}_j(R)$ with $\text{SMS}_j(R) = 0$ for uncovered j ; hence $1 - \text{SMS}(R) = \sum_{j \notin \text{Cov}(R)} w_j + \sum_{j \in \text{Cov}(R)} w_j(1 - \text{SMS}_j(R))$, an exact identity with non-negative terms. (*Computability.*) w_j from fiber-label counts, $\text{SMS}_j(R)$ from row-wise kill rates, $\text{Cov}(R)$ from the checker labels carried by the kill matrix's column index: no equivalence oracle beyond the M_{neq} certification and no ground truth beyond the labels. $\square$

Proof of the cross-zero corollary. If no covered stratum has positive mass, $\text{SMS}(R) = \sum_{j \in \text{Cov}(R)} w_j \text{SMS}_j(R) = 0$. $\square$

Justification of the identifiability remark. Same-stratum membership of a killed mutant's signature is the block-diagonal matrix reread row-wise: a killed member of M_j can only be flagged by checkers with label j , so its nonempty signature certifies the kill and names the fiber; with one exact checker per covered stratum the killed subpopulations separate every covered stratum. Survivor rows are identically zero, hence observationally indistinguishable: the below-window remainder of every covered fiber (window theorem, clause (ii)) and every uncovered fiber produce the same all-pass observation, so survivor attribution is carried by generation-time labels, not by execution outcomes. $\square$

G.8 Detection window: hypotheses, regime, and full proofs

Hypotheses. (*H-a, additive residual budget.*) On the executed tuple at input $x \in D_r$, the residual the checker computes for the mutant satisfies $|\epsilon_r(x; m) - \epsilon_{\text{viol}}(m)| \leq \Delta_r + p\bar{\eta}$: the ideal mutant residual decomposes into the structural violation contribution plus the structure-preservation residual, bounded by Δ_r because the edit perturbs the declared structure, not the discretisation; and the executed tuple aggregates p program executions, each observation carrying noise at most $\bar{\eta}$. The pairwise relations that dominate the catalogue have $p = 2$; the convergence-order relations evaluate $p = 4$ grid

resolutions and their regression verdict additionally carries a design-dependent conditioning constant, so their window bounds are read with the $p = 4$ budget. (*H-b, Lipschitz transfer.*) The violation functional is L_r -Lipschitz in $\varepsilon_{\text{viol}}(m)$; H-b is used only for the dose-response transfer between the nominal-parameter axis and the realized violation axis, not for bounds (i)–(iii), which are stated on the realized axis. Estimability of L_r is assessed per PUT class: linear and quasi-linear kernels admit closed-form or regression estimates, while chaotic and stochastic-structural kernels do not admit a stable L_r on the raw observable, and any dose-response consumption of H-b is restricted to the estimable subset and pre-registered as such. (*H-c, non-degenerate-margin regime.*) Two clauses: (R1) margin dominance, $\mu_r = \varepsilon_{\text{tol}} - \Delta_r > p\bar{\eta}$ strictly; and (R2) ε_{eq} -separation, no executed residual within ε_{eq} of the threshold, making the verdict invariant under pointwise obs-output perturbations of magnitude at most ε_{eq} . R1 is consumed by (i) and (iii) and by the gap theorem’s premise (iii); R2 is consumed by the kill-witness lemma. For unbounded noise the bound $|\eta| \leq \bar{\eta}$ is a concentration-style model assumption and stochastic-case statements hold at the confidence attached to c . (*H-d, magnitude realization.*) The executed tuple realizes the declared magnitude; for uniform parametric violation templates every admissible input realizes it, and unrealized magnitudes only weaken (i), reverting detection to finite-input search, while (ii) and (iii) are unaffected.

Proof of (i). By H-a’s lower side, $\varepsilon_r(x; m) \geq \varepsilon_{\text{viol}}(m) - \Delta_r - p\bar{\eta} > \varepsilon_{\text{tol}}$, so the checker flags the mutant (strict exceedance); by R1 every executed residual of the original program is at most $\Delta_r + p\bar{\eta} < \varepsilon_{\text{tol}}$, so the checker passes the original; the kill predicate’s two conjuncts hold. $\square$

Proof of (ii). By H-a’s upper side, on every executed tuple $\varepsilon_r(x; m) \leq \varepsilon_{\text{viol}}(m) + \Delta_r + p\bar{\eta} < \varepsilon_{\text{tol}}$, so the checker never flags the mutant and the kill predicate’s second conjunct fails; the verdict on the original is not needed. $\square$

Proof of (iii). The lower edge is the contrapositive of (ii); at equality the executed residual is at most ε_{tol} , which under strict exceedance still yields no flag, so killed magnitudes exceed $\varepsilon_{\text{tol}} - \Delta_r - p\bar{\eta}$ strictly. Magnitudes at or beyond $\varepsilon_{\text{crash}}$ violate S4 latency and are excluded from the admitted universe by the crash oracle, so the kill region lies within $(\varepsilon_{\text{tol}} - \Delta_r - p\bar{\eta}, \varepsilon_{\text{crash}})$. $\square$

Proof of the weak-MR remark. $\mu_r < 0$ iff $\Delta_r > \varepsilon_{\text{tol}}$; since Δ_r is a supremum over D_r , inputs exist whose correct-program residual exceeds ε_{tol} (no attainment needed). If validation executes an input whose residual exceeds $\varepsilon_{\text{tol}} + p\bar{\eta}$, the observed residual stays above ε_{tol} under any admissible noise and the flag is guaranteed; inputs at that level exist exactly when $\mu_r < -p\bar{\eta}$. In the band $(\varepsilon_{\text{tol}}, \varepsilon_{\text{tol}} + p\bar{\eta}]$ the flag occurs unless noise masks it, and in all cases validation sampling must reach such inputs at all. A flagged correct program breaks the kill predicate’s pass-on-original conjunct, so the relation contributes no kills and is rejected upstream at MR validation. $\square$

Proof of the stochastic remark. For a pairwise relation ($p = 2$), substituting $\bar{\eta} = c\sigma_{\text{out}}/\sqrt{N} + \eta_{\text{det}}$ into (i) at target magnitude $\varepsilon^\dagger$ and solving for N gives $\sqrt{N} > 2c\sigma_{\text{out}}/(\varepsilon^\dagger - \varepsilon_{\text{tol}} - \Delta_r - 2\eta_{\text{det}})$, provided the denominator is positive; squaring gives the stated bound, which is sufficient with strict inequality; necessity holds under the boundary-attaining noise model used to derive the budget. If the side condition fails, no repeat budget guarantees detection and the remedy is a different observable. $\square$

Structure-fate correspondence. Δ_r places each MR on a structure-preservation axis: $\Delta_r = 0$, exact preservation; $0 < \Delta_r \leq \varepsilon_{\text{tol}}$, approximate preservation (the strong/tolerance MR of the main text); $\Delta_r > \varepsilon_{\text{tol}}$, structure broken (weak MR, the first remark’s $\mu_r < 0$); $\Delta_r(h) \rightarrow 0$ under refinement, asymptotic preservation. The window theorem prices the same Δ_r into two-sided detection bounds: how much violation magnitude a valid relation must convert into a kill and how much it must miss.

H Adjoint Extension Study

This appendix gives the full specification of the confirmatory adjoint extension arm summarised in the main manuscript. The arm exercises the adjoint invariant ψ_6 on three third-party linear-operator libraries, disjoint from the 12 PUTs and the pre-registered 60-cell. Each library carries a real fixed defect in its adjoint code path as a ground-truth anchor:

- **scipy.sparse.linalg.LinearOperator** (linear algebra; BSD-3). Program under test: the scaled-operator adjoint M^H for $M = \alpha A$ with complex scale α . Real anchor: scipy issue #8900 / PR #8962 (the adjoint dropped the conjugate on α , returning αA^H instead of $\bar{\alpha} A^H$).
- **pylops.signalprocessing.FFT** with `real=True` (signal processing; BSD-3). Program under test: the real-FFT adjoint `_rmatvec`; the adjoint identity is the library’s own *dot-test*. Real anchor: pylops issue #268 / PR #292 (commit bd3a919), a wrong scaling of the middle/Nyquist frequency bin that made the adjoint inconsistent with the forward.
- **jax.linear_transpose / vjp** (automatic differentiation; Apache-2.0). Program under test: the autodiff-derived transpose M^T of a linear operator containing an inverse real FFT, $M(x) = \text{irfft}(\text{rfft}(x) \cdot w)$. Real anchor: jax issue #6223 / PR #6232 (commit 112ae41), an `irfft` transpose rule that reversed order with an off-by-one index, so `vjp` silently produced a non-adjoint transpose.

For each subject the MR families are `adj.self` (self-adjointness) and `adj.dual` (scaled/dual-operator transpose). The mutant pool spans adjoint-breaking fault mechanisms (drop-conjugate, sign flip, scale and phase errors, transpose-without-conjugation, Nyquist mis-scaling), with the real fixed defect included among them, plus semantically equivalent mutants (scalar-multiplication reorder, conjugation of a real adjoint output) that a strong MR must *not* kill. Because the pre-fix releases are uninstallable on the host platform (no arm64 wheel), each real defect is reproduced in extracted-diff form on the current release, mirroring the boundary study; ϵ_{tol} is an explicit factor throughout.

Table 16. Adjoint extension study, forward kill.

Substrate	Operator class (real anchor)	Active killed / total	Equiv. survived	Forward SMS
scipy LinearOperator	linear algebra (#8900)	5 / 5	2 / 2	1.00
pylops FFT real	signal processing (#292)	4 / 4	2 / 2	1.00
jax vjp / transpose	automatic differentiation (#6223)	4 / 4	2 / 2	1.00

Across all three substrates the correct program survives, every non-equivalent (active) mutant is killed, including the three real fixed defects, and no semantically equivalent mutant is killed, so the forward SMS is 1.00 with zero false positives on the equivalent set. The scipy #8900 defect is shared with the strong-boundary study in the main manuscript (`adj.self` case) and is not counted as independent corroboration twice; the pylops #292 and jax #6223 defects are unique to this arm. The pools are hand-constructed and low-cardinality, so the arm demonstrates the duality forward rather than measuring a high-adequacy MR set on a discovered fault population (main-paper adjoint-arm reproduction-fidelity threat).

I Real-Defect Evidence Summary

This appendix records only the result-level facts used by the main manuscript’s real-defect evidence slice. It deliberately excludes candidate mining, rejected cases, repository tooling, and benchmark-release narrative. The purpose is auditability for the semantic-mutation argument: kill-rate, semantic-stratum alignment, and real-defect detection are related but distinct constructs.

Table 17. Result-level real-defect evidence summary.

Audit item	Evidence content used in the manuscript	Main-text role
Verified stratum counts	34 verified MR-detectable defects across five meta-patterns: m_{mono} 10, m_{conv} 9, m_{inv} 6, m_{adj} 5, and m_{rev} 4	Shows that the declared semantic strata are not only toy labels
Mutation-phase aggregate	The semantic-mutation track exceeds the closest baseline by +0.101 on mean score (Holm-adjusted $p = 0.046$, Cliff’s $\delta = +0.247$); the remaining pairwise contrasts are not significant	Supports a modest, qualified mutation-score advantage rather than a dominance claim
Real-defect face	The semantic-mutation track detects all 34 tabled real defects; the two generic baseline groups have nonzero detection in 7/34 and 8/34 cases, and the two ablated variants miss 19/34 and 17/34 cases, respectively, with 11 shared misses	Separates real-defect detection from aggregate mutant kill-rate
Non-nesting cases	Four verified non-nesting defects (A-LAPACK-004, A-OPENBLAS-001, B-POCKETFFT-002, and E-ORDINARYDIFFEQ-001) spanning linear algebra, FFT, and ordinary-differential-equation solvers	Supports the semantic-effect fiber view rather than a single scalar hierarchy

This summary is intentionally a result table, not a corpus or benchmark description. The main manuscript uses it only to support construct separation: a relation family can score similarly on aggregate mutant kill-rate while differing sharply on real-defect detection, and different relation families can observe non-nested semantic-effect fibers.

References

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